# VIGNESH SUGUNATHAN

**EMBEDDED FIRMWARE | PRODUCT DEVELOPMENT**

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## PROFESSIONAL SUMMARY

Embedded Systems Engineer with hands-on exposure to the electronics product cycle: board bring-up, firmware development, integration testing and field validation. Built STM32, ESP32 and PIC based systems, including a solar LoRa network with 96.4% packet delivery in outdoor trials and a hazard-detection robot that placed runner-up nationally. Skilled at isolating hardware-firmware faults with oscilloscope, logic analyzer and SWD tools. Aims to help a product team deliver reliable firmware, faster bring-up and validation, and a smooth prototype-to-release path.

## TECHNICAL SKILLS

**Embedded & Firmware:** Embedded C/C++, ARM Cortex-M (STM32F103), ESP32/ESP8266, PIC16F877A, 8051, MSP430, Arduino; Bare-Metal & HAL, GPIO, ADC, PWM, Timers, ISR, DMA, RTC, EEPROM, Low-Power, FSM

**Protocols & Wireless:** UART, SPI, I2C, LoRa (SX1278, 433 MHz), Wi-Fi, Bluetooth/BLE, GSM, GPS (NEO-6M), AES-128, Packet Framing & CRC, RSSI Multi-Hop Routing

**Hardware & Validation:** Board Bring-up & Testing, Sensor/Actuator Interfacing, Motor Control (H-Bridge, BLDC, Stepper), Power-Rail Verification, Fault Isolation, Oscilloscope, Logic Analyzer, DMM, Soldering

**EDA, Tools & Platforms:** KiCad, Proteus, STM32CubeIDE, MPLAB X, Keil, PlatformIO, ST-Link (SWD), Git, Linux, ROS 2, Gazebo, RViz, Python

## PROFESSIONAL EXPERIENCE

**Robotics Instructor** *Sep 2026 – Present*

*NxtWave Institute of Advanced Technologies, Chennai*

- Develop robotics and drone projects on ROS 2 / Ubuntu in Python, simulating robots in Gazebo and visualizing state, sensor streams and transforms in RViz.
- Build IoT and robotics prototypes on ESP32 and STM32, writing and debugging Embedded C/C++ firmware with sensors, motor drivers and Wi-Fi/Bluetooth
- Perform hardware testing and fault isolation across UART, I2C and SPI buses, power rails, wiring and sensor/actuator interfaces.

**AI Robotics Engineer** *Dec 2025 – Aug 2026*

*Centrado Tech Solutions Pvt. Ltd., Chennai*

- **Firmware development:** Wrote Embedded C/C++ for ESP32 / STM32 (Cortex-M): GPIO, ADC, PWM, timers, interrupts and UART/I2C/SPI drivers for
- **Control & sensor fusion:** Implemented real-time sensor fusion and closed-loop motor control (PWM, H-Bridge, BLDC, stepper) for autonomous navig
- **Board bring-up & testing:** Power-rail verification, peripheral validation and board-level functional testing; isolated faults with multimeter, oscilloscope
- **Prototyping & design:** Built harnesses, CAD models and 3D-printed prototypes; supported schematic/PCB reviews, tuned low-latency firmware and del

## TOP ENGINEERING PROJECTS

**Decentralized Solar-Powered Multi-Node LoRa Mesh Emergency Network** *Field-tested*

*LoRa SX1278 (433 MHz) | ESP32 | Arduino Nano | ESP8266 | GSM | NEO-6M GPS | AES-128 | Solar Power Mgmt*

- Architected an off-grid, multi-node network with dynamic RSSI-based multi-hop routing, GPS location reporting, GSM alerting and AES-128 encrypted payload
- Achieved ~96.4% packet delivery over 500 transmissions at up to 200 m in outdoor field tests; engineered solar harvesting and power management for continuou

**Autonomous Surveillance & Hazard Detection Robot** *2025 | Runner-Up, SRM National Robotics Hackathon*

*ESP32 | Arduino Uno | MQ-2 Gas | HC-SR04 | L298N H-Bridge | Voice & IR Control | Blynk | Bluetooth SPP*

- Built a 4WD mechatronic platform with ultrasonic obstacle avoidance, line tracking, voice/IR/Bluetooth control, PWM motor speed control, and real-time toxic-

**STM32 Real-Time Environmental Monitor & I2C OLED Driver** *STM32F103C8T6 (Cortex-M3 @ 72 MHz)*

*Embedded C | STM32CubeIDE | ST-Link V2 SWD | FTDI UART | I2C | DS1307 RTC | DHT11 | SSD1306 OLED*

- Performed bare-metal bring-up (HSE -> PLL 72 MHz clock tree, I2C fast-mode peripheral, timers/interrupts) and wrote register-level drivers for OLED display,
- Validated bus behavior and sensor acquisition through SWD in-circuit debugging, logic-analyzer decoding and a non-blocking 115200-baud UART diagnostic st

**Microwave Oven Automation System (Embedded Systems Internship, Emertxe)** *Jan – Feb 2025*

*PIC16F877A | Embedded C | MPLAB X (XC8) | PICSimLab | GPIO | ADC | Timers | Interrupts | EEPROM | Matrix Keypad*

## EDUCATION

**B.E. – Electronics and Communication Engineering** *2022 – 2026*

*Agni College of Technology, Chennai (Anna University)* **CGPA: 8.2 / 10**

## ACHIEVEMENTS & PUBLICATIONS

- **Best Research Paper Award, ICRIDET 2024 (IEEE)** for Real-time Oil Spill Detection Using Deep Learning and CNNs; second IEEE paper on dent da
- **Runner-Up, SRM National Robotics Hackathon (2025)** | **NPTEL Embedded Systems and Design (2024)**